

## Assignment 1.

$\{C\}$  is correct answer.

Because  $X_i$  Along the direction of  $a_i$ .

## Assignment 2:

DH table of the PUMA 560 6-axis robotic arm.

| $i$ | $\alpha_{i-1}$ | $a_{i-1}$ | $d_i$ | $\theta_i$ |
|-----|----------------|-----------|-------|------------|
| 1   | 0              | 0         | 0     | 0          |
| 2   | -90            | 0         | 220   | 0          |
| 3   | 0              | 430       | -90   | 0          |
| 4   | -90            | 430       | 0     | 0          |
| 5   | 90             | 0         | 0     | 0          |
| 6   | -90            | 0         | 0     | 0          |

## Assignment 3:

$$\theta_1 = 15^\circ \quad \theta_2 = 40^\circ \quad \theta_3 = -50^\circ \quad \theta_4 = 30^\circ \quad \theta_5 = 70^\circ \quad \theta_6 = 25^\circ$$

$$1) {}^1_2T = \begin{bmatrix} {}^1_2R & {}^1P_{2org} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_1 \\ s\theta_2 c\alpha_1 & c\theta_2 c\alpha_1 & -s\alpha_1 & -s\alpha_1 d_2 \\ s\theta_2 s\alpha_1 & c\theta_2 s\alpha_1 & c\alpha_1 & c\alpha_1 d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.766 & 0.6428 & 0 & 0 \\ 0 & 0 & 1 & 220 \\ 0.6428 & -0.766 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$2) {}^0_1T = \begin{bmatrix} {}^0_1R & {}^0P_{1org} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_0 \\ s\theta_1 c\alpha_0 & c\theta_1 c\alpha_0 & -s\alpha_0 & -s\alpha_0 d_1 \\ s\theta_1 s\alpha_0 & c\theta_1 s\alpha_0 & c\alpha_0 & c\alpha_0 d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.9659 & -0.2588 & 0 & 0 \\ 0.2588 & 0.9659 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} {}^2_3R & {}^2P_{3org} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 c\alpha_2 & c\theta_3 c\alpha_2 & -s\alpha_2 & -s\alpha_2 d_3 \\ s\theta_3 s\alpha_2 & c\theta_3 s\alpha_2 & c\alpha_2 & c\alpha_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.6428 & 0.766 & 0 & 430 \\ -0.766 & 0.6428 & 0 & 0 \\ 0 & 0 & 1 & -90 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} {}^3_4R & {}^3P_{4org} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ s\theta_4 c\alpha_3 & c\theta_4 c\alpha_3 & -s\alpha_3 & -s\alpha_3 d_4 \\ s\theta_4 s\alpha_3 & c\theta_4 s\alpha_3 & c\alpha_3 & c\alpha_3 d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.8660 & -0.5 & 0 & 430 \\ 0 & 0 & 1 & 0 \\ -0.5 & -0.8660 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$So {}^0_4T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T = \begin{bmatrix} 0.1294 & 0.2241 & 0.9658 & 284.5 \\ -0.4820 & -0.8245 & 0.2588 & 210.8 \\ 0.8660 & -0.4599 & 0 & 706.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0P_{4org} = \begin{bmatrix} 284.5 \\ 210.8 \\ 706.4 \end{bmatrix}$$

$${}^0P_4 \text{ org} = \begin{bmatrix} 284.5 \\ 210.8 \\ 706.4 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & a_4 \\ s\theta_5 c\alpha_4 & c\theta_5 c\alpha_4 & -s\alpha_4 & -s\alpha_4 d_5 \\ s\theta_5 s\alpha_4 & c\theta_5 s\alpha_4 & c\alpha_4 & c\alpha_4 d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.3240 & -0.9397 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0.9397 & 0.3240 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & a_6 \\ s\theta_6 c\alpha_5 & c\theta_6 c\alpha_5 & -s\alpha_5 & -s\alpha_5 d_6 \\ s\theta_6 s\alpha_5 & c\theta_6 s\alpha_5 & c\alpha_5 & c\alpha_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.9063 & -0.426 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -0.4226 & -0.9063 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6T = {}^0_4T \times {}^4_5T \times {}^5_6T$$

$$= \begin{bmatrix} 0.8225 & -0.3825 & 0.1713 & 284.5 \\ 0.2203 & -0.1028 & 0.5376 & 210.8 \\ 0.0430 & -0.5716 & -0.8137 & 706.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$